

SAI SATWIK BIKUMANDLA

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PROFESSIONAL SUMMARY

Software engineer who ships complete systems - not just components. Built and deployed a production-grade, version-controlled resume platform (Next.js · MongoDB · LaTeX) made for the real world, a real-time YOLO-based sports tracking pipeline, and a cloud-secured relational database on Google Cloud SQL. Equally fluent in frontend, backend, and ML layers, with a track record of taking ambiguous problems from idea to working software.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, C++
- **Web & Backend:** React.js, Next.js (App Router), Node.js, Flask, REST APIs, NextAuth, JWT, HTML/CSS
- **Databases:** MySQL, MongoDB, Google Cloud SQL, schema design, normalization, query optimization
- **ML & Vision:** TensorFlow/Keras, YOLO, LSTM/RNN, TF-IDF, cosine similarity, NLP preprocessing
- **Tools & Cloud:** Git, Google Cloud Platform, Docker, LaTeX, MATLAB, Gemini API

EDUCATION

University at Albany, State University of New York (SUNY)

Aug 2024 - May 2026

Master of Science, Computer Science

Albany, NY

- **Coursework:** Operating Systems, Computer Security, Algorithms & Data Structures, Database Systems, Artificial Intelligence, Computer Vision, Probability & Computing

Sri Indu College of Engineering and Technology (JNTUH)

Aug 2020 - Apr 2024

Bachelor of Technology, Computer Science & Information Technology

Hyderabad, India

- **Coursework:** Machine Learning, AI, Compiler Construction, Computer Networks, Software Engineering, Software Testing Methodologies

PROJECTS

LaTeX Resume Builder - Version-Controlled Resume Platform

2025

Independent Development

- Designed and shipped a full-stack SaaS-style web application (Next.js, MongoDB, NextAuth) giving job seekers a GitHub-inspired resume workflow: multiple named repositories, commit snapshots, full version history, and rollback to any prior state - eliminating the problem of scattered, inconsistent resume files.
- Engineered a dual-render pipeline: an HTML live preview for zero-latency editing feedback alongside a server-side LaTeX compilation endpoint that exports polished, print-ready PDFs from any historical commit.
- Modeled resume versions as append-only, immutable snapshots (User → ResumeRepository → ResumeVersion), enforcing audit-trail integrity while keeping the UX approachable for non-technical users across 4 purpose-built pages.

Movie Recommendation System - Content-Based NLP Engine

2025

Independent Development

- Built a content-based recommender over a dataset of 5,000+ films using TF-IDF vectorization and cosine similarity; applied a full NLP preprocessing pipeline (tokenization, lemmatization, stop-word removal) to improve feature signal quality.
- Evaluated recommendation relevance using Precision@K across K = 5, 10, and 20, enabling data-driven tuning of the similarity threshold and surfacing measurable lift over a random baseline.

Tennis Ball Detection & Tracking System

2025

University at Albany - Computer Vision

- Integrated Roboflow's pre-trained tennis ball detection model via REST API and built two production-quality tracking systems on top: a Basic Model with anti-jump filtering, exponential smoothing ($\alpha=0.85$), and multi-frame velocity extrapolation; and an Advanced Model adding court-side A/B classification, perspective-aware overlays, and configurable color schemes per court surface type.
- Implemented SORT (Simple Online and Realtime Tracking) for multi-object tracking with custom bounding-box conversion, per-track trajectory trails (50-frame memory), and frame-by-frame annotation overlays displaying Ball ID, detection count, and active track count.
- Built a TrackingAnalytics engine that generates structured JSON output per session - capturing velocity peaks, total trajectory distance, side-time distribution, detection success rate, and position maps - exported alongside processed video.
- Deployed as an interactive Gradio UI with model selection (Basic vs. Advanced), configurable start/end time segments, court color scheme selection, live analytics summary, session run history, and ZIP download of processed video + analytics JSON.

MyDishDB - Cloud-Deployed Relational Database System

2025

University at Albany - Database Systems

- Modeled and implemented a fully normalized MySQL schema (E-R → DDL) spanning 10+ entities: recipes, users, ingredients, categories, equipment, and dietary tags - enforced via CHECK constraints, triggers, and referential integrity rules.

- Deployed on Google Cloud SQL with role-based access control and JWT-authenticated backend; cut query response time by optimizing critical paths with indexed multi-table JOINS and UNION-based aggregation queries.

Next-Word Predictor - LSTM Language Model

2025

University at Albany - Artificial Intelligence

- Trained an LSTM language model (Embedding → LSTM 150 units → Softmax Dense) in TensorFlow/Keras on 2,000 IMDB reviews; built the full data pipeline - tokenization, sliding-window sequencing (length 20), padding, and one-hot encoding - over a 20,000-token corpus with an 80/20 train-validation split.
- Implemented checkpoint-based model persistence: training runs once and subsequent sessions load saved weights instantly, enabling iterative experimentation without redundant computation.

Web Data Detection & Monitoring System

2025

University at Albany - Academic Project

- Architected an automated web monitoring pipeline that tracks and extracts structured data from dynamic sources on a scheduled basis, with configurable alerting when target conditions are met.
- Implemented a modular, fault-tolerant architecture with structured logging and layered exception handling - designed to remain operational without manual intervention across extended monitoring windows.

Web Prototype Generator - AI-Assisted Code Generation Tool

2024

University at Albany - Team Project (3 members)

- Led the Gemini API integration layer, building the prompt engineering pipeline that converts natural-language client briefs into functional, responsive HTML/CSS/JS prototypes with auto-embedded Unsplash imagery; backend service layer built with Flask.
- Introduced in-source AI-attribution tagging and documentation standards adopted across the team codebase, ensuring transparency on human vs. AI-generated contributions.

PROFESSIONAL EXPERIENCE

Rethink UX | *Web Developer Intern*

Oct 2022 - Dec 2022

- Selected for a competitive engineering internship while simultaneously carrying a full undergraduate course load - one of the earliest professional engagements in a cohort of CS peers.
- Took sole ownership of 3 client-facing web feature modules from requirements to deployment; independently designed and implemented a role-based access control system, a pattern later reused across the product as a standard security layer.
- Identified gaps in team knowledge transfer and self-initiated the creation of technical documentation; the docs were adopted into the standard onboarding workflow, reducing ramp-up friction for engineers joining after the internship ended.
- Earned consistent code review approvals from senior engineers within the first two weeks - demonstrating the ability to write production-quality code under professional standards without a grace period.

StudyExperts | *Developer Advocate*

Jun 2022 - Aug 2022

- Joined as the technical bridge between engineering and users during undergraduate Year 2 - tasked with translating complex product behavior into clear, actionable content for a developer audience.
- Researched, wrote, and published a structured library of technical tutorials and knowledge-base articles that became the primary self-service resource for the platform, measurably reducing the volume of repeated support escalations to the engineering team.
- Operated as an informal QA layer: systematically reproduced 10+ edge-case defects, documented them with full reproduction steps and environment context, and delivered reports precise enough for engineers to act on immediately - without follow-up clarification cycles.
- Demonstrated that strong written communication and deep product understanding are force multipliers for small engineering teams - a lesson that now directly informs how I document and advocate for my own projects.

LANGUAGES

- Telugu (Native)
- Hindi (Fluent)
- English (Fluent)
- German (Actively Learning)